

CLOUD LOGIN THREAT DETECTION SYSTEM

SOC Incident Investigation Report

Incident: Multiple Failed Login Attempts
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Microsoft Azure | Microsoft Sentinel | Microsoft Defender | Microsoft Entra ID

Contents

- 1. Executive Summary 2
- 2. Incident Details 2
- 3. Detection Logic 2
- 4. Evidence and Investigation..... 2
 - 4.1 Azure Resource and Workspace..... 3
 - 4.2 Log Analytics Workspace Overview 3
 - 4.3 KQL Query History 4
 - 4.4 Failed Login Query Result 4
 - 4.5 Microsoft Sentinel Analytics Rule 5
 - 4.6 Analytics Rule Description 5
 - 4.7 Account Lockout During Attack Simulation..... 6
 - 4.8 Microsoft Defender and Sentinel Monitoring Overview 7
 - 4.9 Active Incidents Dashboard 8
 - 4.10 Microsoft Defender Incident Queue..... 9
 - 4.11 Incident and Alert Summary10
 - 4.12 Alert Description and Rule Evidence.....11
 - 4.13 Alert Metadata13
 - 4.14 Alert Classification State15
- 5. Findings and Assessment16
- 6. MITRE ATT&CK Mapping16
- 7. Response Actions16
- 8. Recommendations16
- 9. Conclusion.....17

1. Executive Summary

This report documents a controlled cloud identity security test in which repeated failed Microsoft Entra ID sign-in attempts were generated against a test account. Authentication events were collected in the CloudThreat-LAW Log Analytics workspace and analysed with KQL. A custom Microsoft Sentinel scheduled analytics rule detected five failed attempts and created a medium-severity incident in Microsoft Defender.

The investigation confirmed that the activity was part of the authorised lab simulation. The test account was temporarily locked, no successful unauthorised sign-in was observed, and no evidence of data exposure or privilege escalation was identified.

2. Incident Details

Incident ID	20
Incident title	Multiple Failed Login Attempts
Severity	Medium
Category	Credential Access
Status	New / Active
Detection source	Microsoft Sentinel scheduled detection
Workspace	CloudThreat-LAW
Affected identity	testuser@alekhyapanguluri23gmail.onmicrosoft.com
Observed threshold	5 failed attempts

3. Detection Logic

The custom rule searches Microsoft Entra ID sign-in records for unsuccessful authentication events, groups the events by user, and returns accounts with five or more failures during the configured one-hour period.

SigninLogs

| where TimeGenerated > ago(1h)

| where ResultType != 0

| summarize FailedAttempts = count() by UserPrincipalName

| where FailedAttempts >= 5

| order by FailedAttempts desc

4. Evidence and Investigation

The following evidence is presented in investigation order. Each heading now matches the screen shown directly below it.

4.1 Azure Resource and Workspace

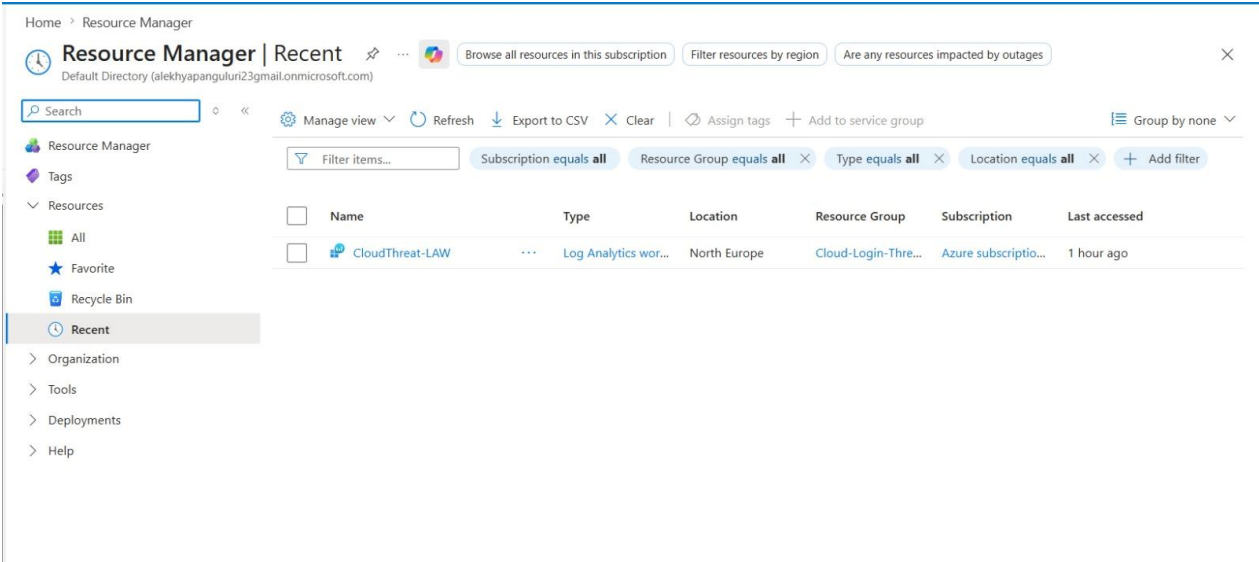


Figure 1. Azure Resource and Workspace

The Azure Resource Manager view confirms that the CloudThreat-LAW Log Analytics workspace was deployed in North Europe and associated with the cloud login threat-detection resource group. This workspace provides the central location for storing and querying identity logs.

4.2 Log Analytics Workspace Overview

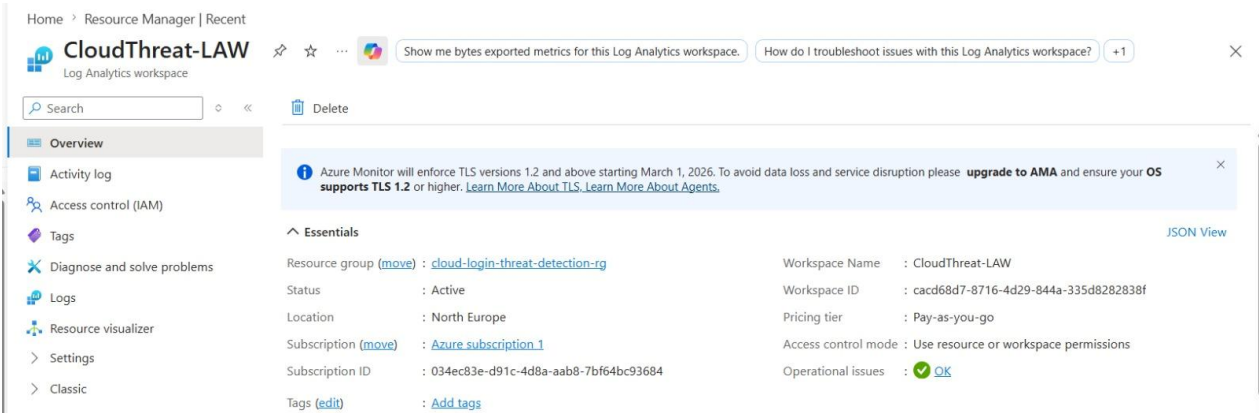


Figure 2. Log Analytics Workspace Overview

The workspace overview shows CloudThreat-LAW in an active state. The operational status confirms that the workspace was available to receive Microsoft Entra ID sign-in data and support Microsoft Sentinel analytics.

4.3 KQL Query History

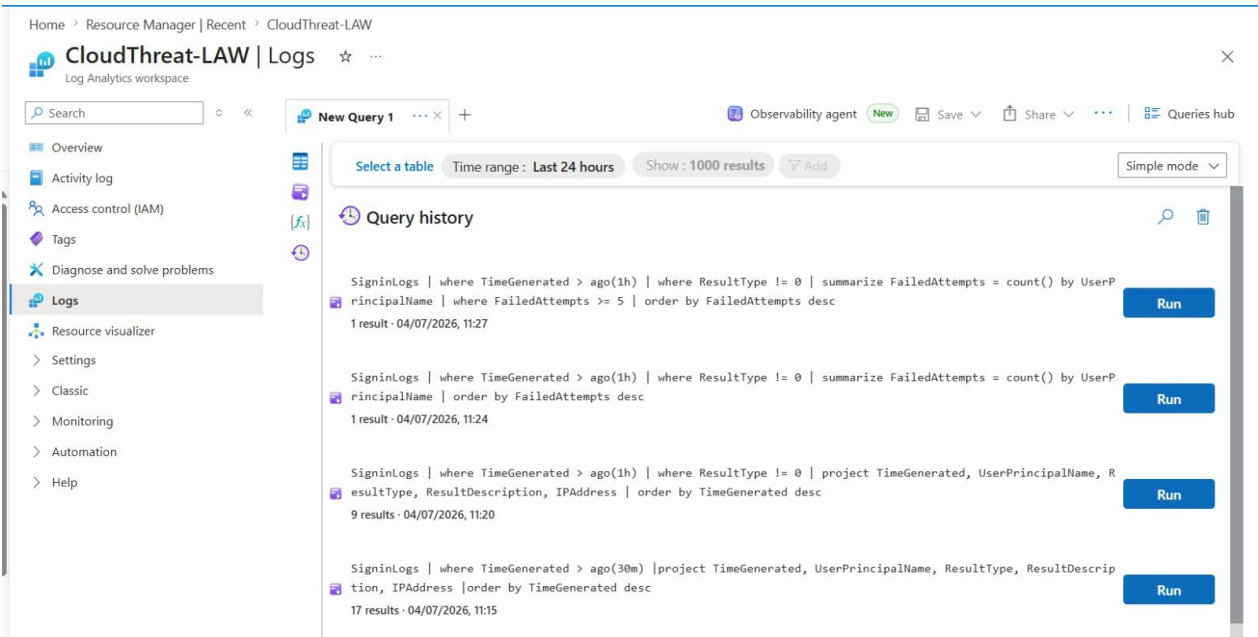


Figure 3. KQL Query History

The query history records the KQL statements used during the investigation. These queries reviewed recent sign-ins, filtered failed results, projected relevant fields, and counted failed attempts by user.

4.4 Failed Login Query Result

Related events	
Query results	
View query	
1 item	
Customize columns	
UserPrincipalName	FailedAttempts
testuser@alekhy...	5
UserPrincipalName	testuser@alekhyapanguluri23gmail.onmicrosoft.com
FailedAttempts	5

Figure 4. Failed Login Query Result

The related-events result identifies the test user and shows FailedAttempts = 5. This is the key evidence that satisfied the analytics-rule threshold and caused the alert to be generated.

4.5 Microsoft Sentinel Analytics Rule

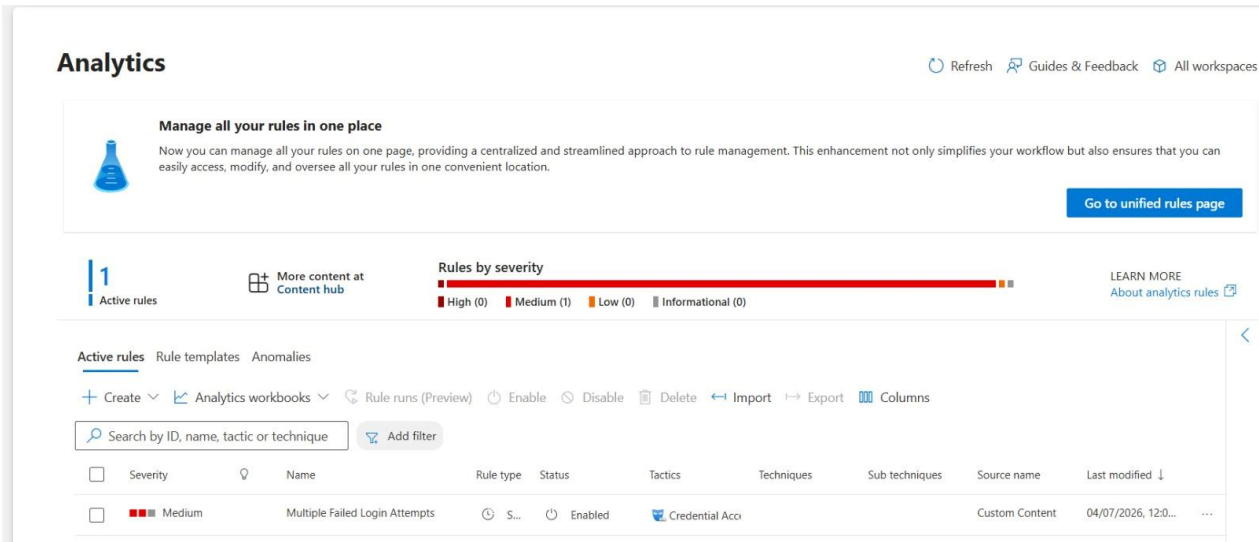


Figure 5. Microsoft Sentinel Analytics Rule

The Analytics page shows one active custom rule named Multiple Failed Login Attempts. The rule is enabled, assigned medium severity, and mapped to the Credential Access tactic.

4.6 Analytics Rule Description

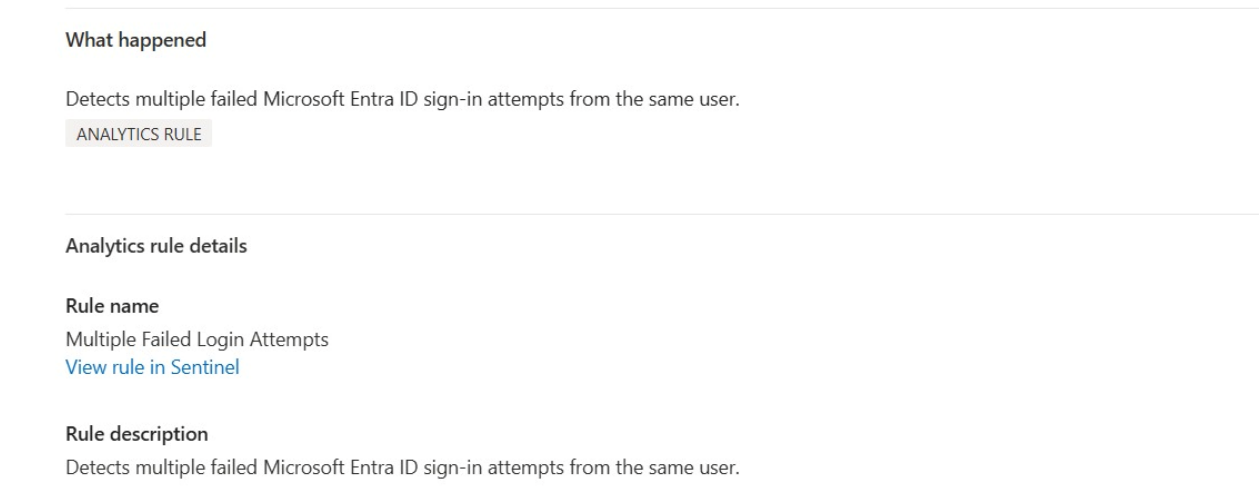


Figure 6. Analytics Rule Description

The rule details state that the detection is designed to identify multiple failed Microsoft Entra ID sign-in attempts from the same user. This confirms that the generated incident matches the intended detection use case.

4.7 Account Lockout During Attack Simulation

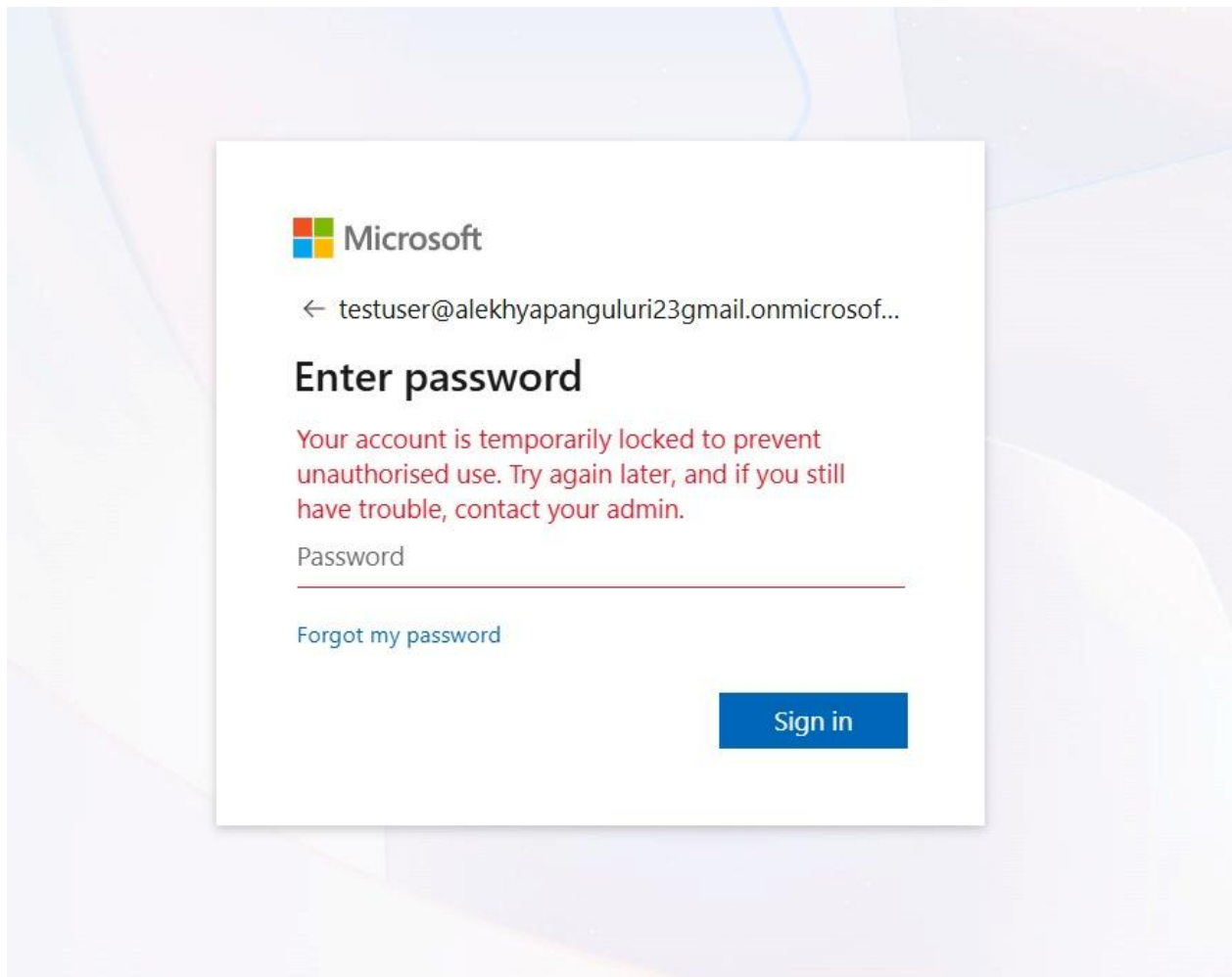


Figure 7. Account Lockout During Attack Simulation

The test account was temporarily locked after repeated incorrect password submissions. This demonstrates the simulated password-guessing activity and provides supporting evidence that Microsoft Entra ID protection controls responded to the failures.

4.8 Microsoft Defender and Sentinel Monitoring Overview

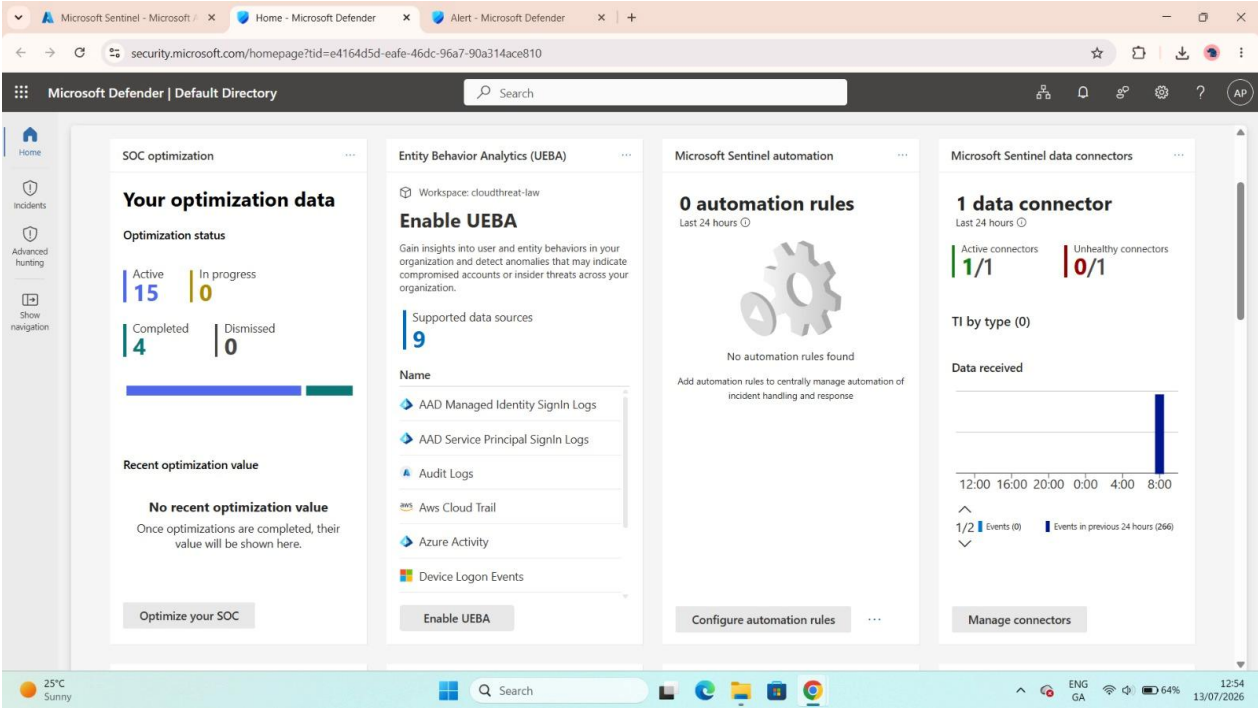


Figure 8. Microsoft Defender and Sentinel Monitoring Overview

The Microsoft Defender home page shows Sentinel monitoring information, including an active data connector and received event data. This confirms that the identity data source was connected and sending telemetry.

4.9 Active Incidents Dashboard

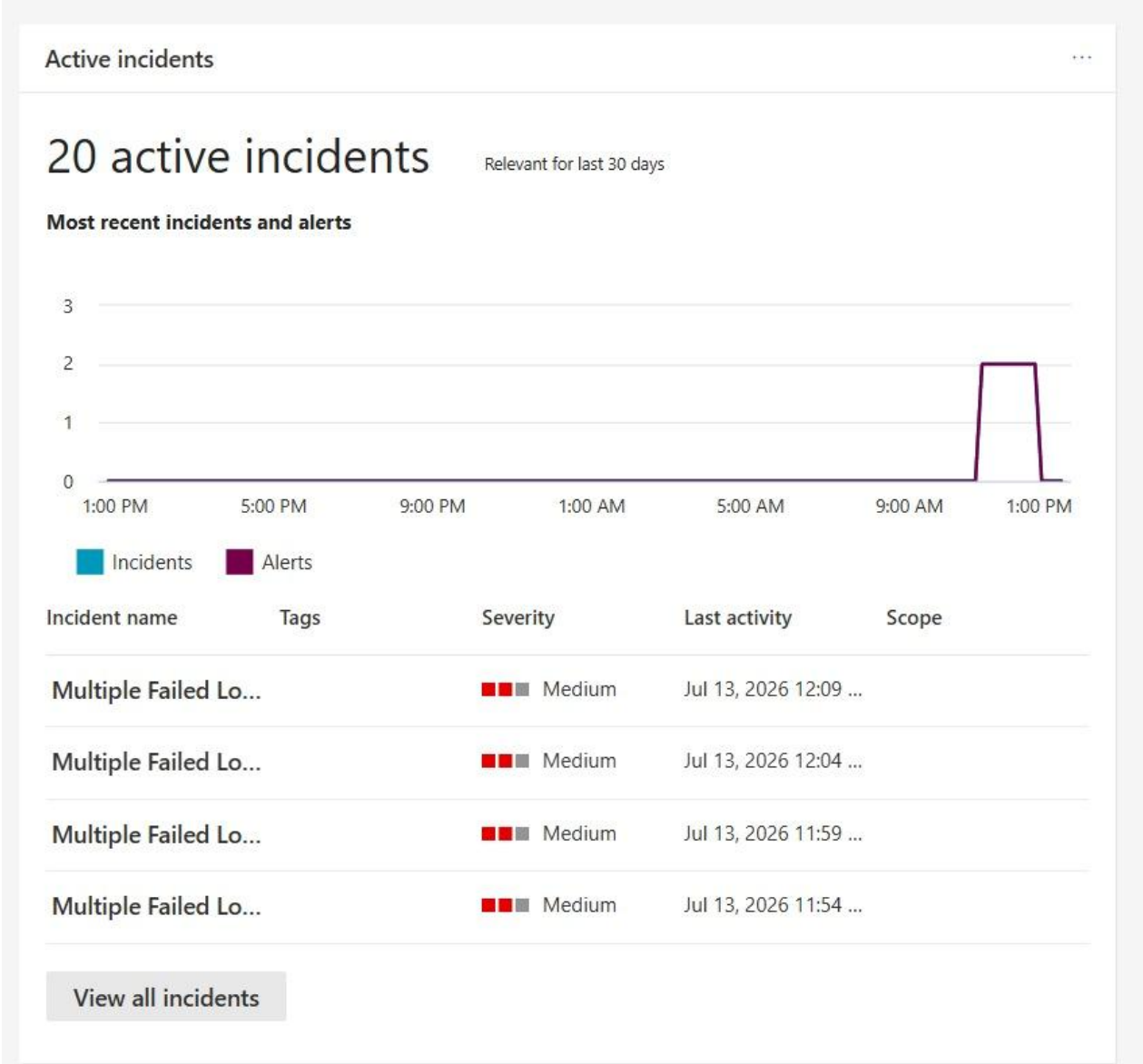
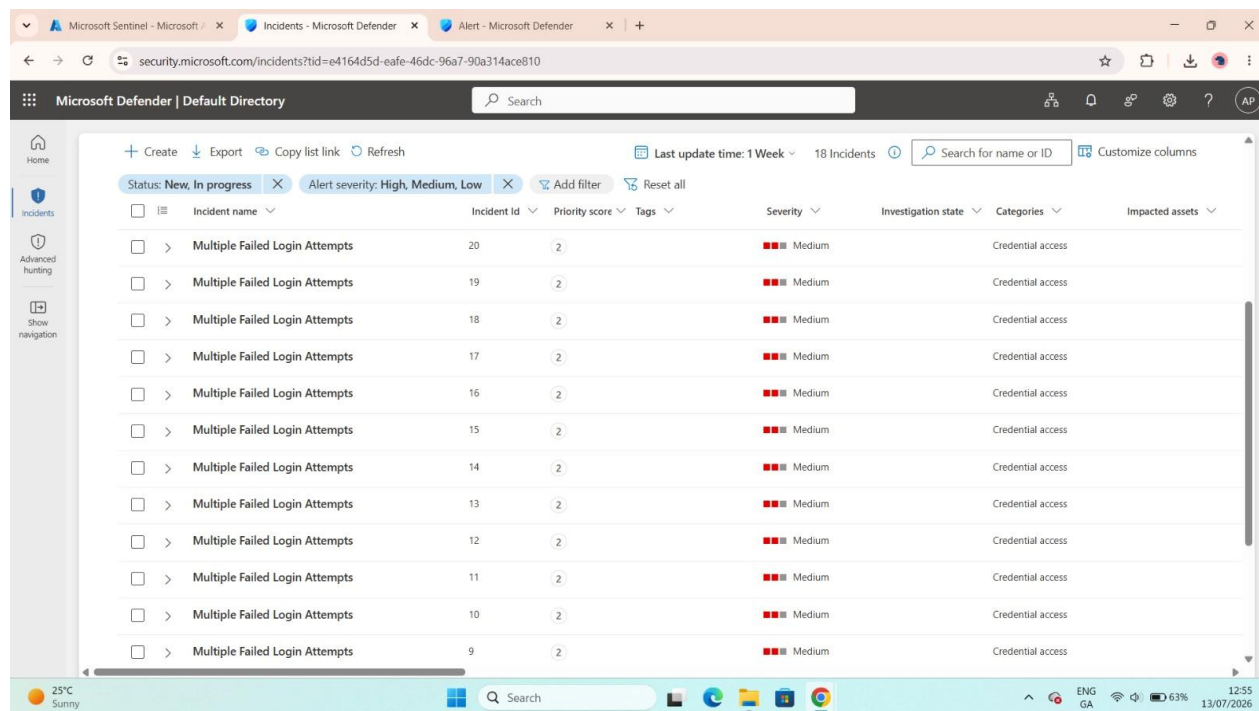


Figure 9. Active Incidents Dashboard

The Active incidents widget displays several recent Multiple Failed Login Attempts incidents with medium severity. Repeated entries occurred because the scheduled rule continued to evaluate matching events during its query period.

4.10 Microsoft Defender Incident Queue



Incident name	Incident ID	Priority score	Tags	Severity	Investigation state	Categories	Impacted assets
Multiple Failed Login Attempts	20	2		Medium		Credential access	
Multiple Failed Login Attempts	19	2		Medium		Credential access	
Multiple Failed Login Attempts	18	2		Medium		Credential access	
Multiple Failed Login Attempts	17	2		Medium		Credential access	
Multiple Failed Login Attempts	16	2		Medium		Credential access	
Multiple Failed Login Attempts	15	2		Medium		Credential access	
Multiple Failed Login Attempts	14	2		Medium		Credential access	
Multiple Failed Login Attempts	13	2		Medium		Credential access	
Multiple Failed Login Attempts	12	2		Medium		Credential access	
Multiple Failed Login Attempts	11	2		Medium		Credential access	
Multiple Failed Login Attempts	10	2		Medium		Credential access	
Multiple Failed Login Attempts	9	2		Medium		Credential access	

Figure 10. Microsoft Defender Incident Queue

The incident queue lists the generated incidents, including Incident ID 20. Each incident is categorised as Credential Access and assigned medium severity, allowing a SOC analyst to prioritise and investigate the activity.

4.11 Incident and Alert Summary

The screenshot shows a web interface for an incident and alert summary. At the top, there's a title bar with a close button (X). Below it, a red lightning bolt icon is next to the title "Multiple Failed Login Attempts". To the right of the title, there are three status indicators: "Medium" (represented by two red squares and one grey square), "Unknown" (represented by a grey circle), and "New" (represented by a blue circle). Below the title, there are two action buttons: "Manage alert" (with a pencil icon) and "Move alert to another incident" (with a shield icon). Below these buttons, there's a link "View rule in Sentinel" and a description "sign-in attempts from the same user." Below this, there's a section titled "Incident details" with an expand/collapse arrow. Under "Incident details", there are two sub-sections: "Incident" and "Incident severity". The "Incident" sub-section shows the text "Multiple Failed Login Attempts". The "Incident severity" sub-section shows "Medium" with two red squares and one grey square. At the bottom, there's a table with five columns: "Active alerts", "Devices", "Users", "Mailboxes", and "Apps". The values in the table are: "Active alerts" (1/1), "Devices" (0), "Users" (0), "Mailboxes" (0), and "Apps" (0).

Active alerts	Devices	Users	Mailboxes	Apps
1/1	0	0	0	0

Figure 11. Incident and Alert Summary

The alert panel links the Multiple Failed Login Attempts alert to its incident and shows one active alert. The panel also confirms medium incident severity and provides the starting point for investigation.

4.12 Alert Description and Rule Evidence



Multiple Failed Login Attempts

Medium

Unknown

New

 Manage alert

 Move alert to another incident

Workspace

cloudthreat-law

Evidence

No related evidence was found

Alert description

Detects multiple failed Microsoft Entra ID sign-in attempts from the same user.

Analytics rule details

Rule name	Rule description
Multiple Failed Login Attempts	Detects multiple failed Microsoft Entra ID

Figure 12. Alert Description and Rule Evidence

The alert details identify the CloudThreat-LAW workspace and display the description of the custom analytics rule. No separate related evidence entity was created, but the alert remains supported by the query results and incident data.

4.13 Alert Metadata



Multiple Failed Login Attempts

Medium

Unknown

New

 Manage alert

 Move alert to another incident

sn51c5a84e-5c2d-4cdd-9c3d-11c01759443d

Alert ID

sn51c5a84e-5c2d-4cdd-9c3d-11c01759443d

Categories

Credential Access

Detection source

Scheduled detection

Service source

Microsoft Sentinel

Detection technology

-

Generated on

Jul 13, 2026 12:15:10 PM

First activity

Jul 13, 2026 11:09:16

Last activity

Jul 13, 2026 12:09:16

Figure 13. Alert Metadata

The metadata view records the alert ID, Credential Access category, scheduled detection source, Microsoft Sentinel service source, generation time, and activity period. These fields establish the technical provenance and timeline of the alert.

4.14 Alert Classification State



Multiple Failed Login Attempts

Medium

Unknown

New

 Manage alert

 Move alert to another incident

INSIGHT

Quickly classify this alert

Classify alerts to improve alert accuracy and get more insights about threats to your organization.

Classify alert

Alert state

Classification

Assigned to

Not Set

Unassigned

[Set Classification](#)

Figure 14. Alert Classification State

The alert was initially new, unassigned, and not classified. In a production SOC workflow, an analyst would assign ownership, validate the activity, and classify it as a benign positive because this event was generated intentionally in a controlled lab.

5. Findings and Assessment

- Five failed Microsoft Entra ID sign-in attempts were recorded for the same test identity.
- The custom scheduled analytics rule operated as designed and generated medium-severity incidents.
- The test account was temporarily locked after repeated unsuccessful authentication attempts.
- No successful unauthorised authentication or evidence of account compromise was identified.
- Repeated incidents were caused by the rule repeatedly evaluating the same matching events within the configured lookback period.

6. MITRE ATT&CK Mapping

Tactic: Credential Access (TA0006)

Technique: Brute Force (T1110)

The activity is mapped to Brute Force because repeated password attempts were made against a cloud user identity. In this lab the attempts were authorised and simulated, but the behaviour resembles password guessing that a threat actor could use to obtain valid credentials.

7. Response Actions

- Reviewed the incident and alert in Microsoft Defender.
- Confirmed the affected identity and failed-attempt count using KQL.
- Verified that the analytics rule and data connector were active.
- Confirmed account lockout and checked for successful authentication.
- Documented the incident as an authorised test and retained screenshots as evidence.

8. Recommendations

- Require multi-factor authentication for all users.
- Apply Conditional Access policies based on sign-in risk, location, and device compliance.
- Tune the analytics rule to reduce duplicate incidents by adjusting query frequency, lookback period, grouping, or suppression.
- Map the user entity in the analytics rule to improve investigation context.
- Create an automation rule or playbook for assignment, notification, and enrichment.
- Monitor source IP addresses, geographic location, result descriptions, and any successful sign-in following repeated failures.

9. Conclusion

The lab successfully demonstrated an end-to-end cloud identity threat-detection workflow. Microsoft Entra ID generated authentication telemetry, Log Analytics stored and queried the events, Microsoft Sentinel applied the custom KQL rule, and Microsoft Defender presented the resulting incidents for investigation. The evidence confirms that the detection rule worked as intended and that the activity did not result in a successful compromise.